

Objectives:

The objective of this lab session is to introduce students to the Linux command-line interface (CLI) and help them become comfortable with its basic usage. Through hands-on practice, students will learn to perform essential file and directory management tasks, record and review their own terminal sessions, and appreciate how the CLI can often be faster, more efficient, and more powerful than graphical interfaces. Although we're primarily using Ubuntu, the commands explored in this session are fundamental to all Unix-like systems, including macOS, making the skills learned here widely applicable.

Program Learning Outcomes (PLOs):

- PLO1: Academic Education
- PLO2: Knowledge for Solving Computing Problems
- PLO3: Problem Analysis
- PLO4: Design/Development of Solutions
- **PLO5: Modern Tool Usage**
- PLO6: Individual and Team Work
- PLO7: Communication
- PLO8: Computing Professionalism and Society
- PLO9: Ethics
- PLO10: Life-long Learning

CLO–PLO Mapping with BT Levels

CLO	Learning Outcome	BT Level	Mappe d PLOs
CL01	Practice Linux commands and utilities to gain proficiency with the command-line interface and use it effectively for system management tasks, basic scripting, software installation, and other essential operations in a Linux environment.	A5	PL05

Practice Exercises

Before we begin today's graded lab exercise:

- Open your **terminal** as we start the session.
- Follow along and replicate everything I do and try to internalize the commands and concepts.
- This will prepare you for today's **graded exercise**.

We will practice following commands today and our graded task will also be based upon them ;

`echo, pwd, whoami, cd, ls, clear, touch, tee, mkdir, mkdir -p, rm, rm -r,`

We will learn some useful keyboard shortcuts which will save us a lot of time while playing around with the Linux commands.

We will learn how to **record terminal activity into a file** so that you can review it later.

We might briefly explore the file hierarchy in linux so we can easily navigate through it.

Graded Exercise

- Open your terminal and **start recording your session**. Name the recording file using your own name (e.g; Ali-Hafeez.txt).
- Display a short introduction message containing your name and roll number on the terminal.
- Check your current working directory. By default, you should be in your home directory.
- Display your username and your machine's hostname.
- Change your current directory to the Desktop and verify the change and list the contents of your Desktop directory.
- **Using a single command** create a hierarchy of nested directories with the following structure:
BNU/SCIT/CS/SEM-V/OS-LAB/Name-of-Student
(You'll name the last directory by your own name)
- Use **ls-R** command to list contents recursively and confirm that the nested directories were created successfully.
- Change your current directory to the **Name-of-Student** directory you just created and again check your current working directory.
- Create **40 text files**. The name of each file should start with your Name (e.g; Ali-1.txt)
(I'm expecting you to use a single command for this purpose)
- Verify that the 40 text files were created successfully by listing the contents within your current

directory.

- Write a short text string (containing your name and roll number) inside all 40 files.
- Display the contents of any 5 random files out of these 40. (Can be done with a single command.)
- Remove all 40 text files in a single step.
- Display a completion message that includes your roll number.
- **Exit the recording session.**

Submission Guidelines

- Take screenshots of your work and submit a pdf document of those screenshots.

